



ICT BANGLADESH
Invention for Next Generation

PRODUCT PROPOSAL
University Management System

Prepared for
AI-Powered Software Engineering Batch 26/3
ICT Bangladesh

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Version
1.0 — Final Draft

Platform
Web/Mobile App + REST API

Target
Universities

1. INTRODUCTION

This document is the official final project brief issued by **ICT Bangladesh** to students of the AI-Powered Software Engineering batch 26/3. Students are required to read this proposal carefully and submit a fully working implementation based on the specifications defined here.

Universities today operate across a web of disconnected systems. Attendance is kept in spreadsheets. Exam results go out by email. Fee records live in a separate finance tool that nobody outside the accounts department can access. Students do not know their attendance percentage until it is too late. Parents find out about missing fees when a notice arrives in the post.

ICT Education consolidates all of this into a single, well-structured platform. The goal is not to add another portal — it is to replace the patchwork of workarounds with one coherent system built for the people who use it every day.

2. PROJECT GRADING STRUCTURE

This project will be evaluated out of a total of 100 marks, divided across two components:

Component	Marks	Description
Main Project Deliverable	80	Fully functional API + Web Application as specified in this proposal.
Assignment & Attendance	20	Interim assignments (10 marks) + session attendance and participation (10 marks).
Total	100	Pass mark: 50. Distinction: 80 and above.

Project (80 marks) — breakdown:

- Working REST API with all specified endpoints — 30 marks
- Web application with complete UI and role-based access — 30 marks
- Database design, relationships, and data integrity — 15 marks
- Code quality, documentation, and deployment — 5 marks (Optional)

Assignment & Attendance (20 marks) — breakdown:

- Session attendance (minimum 80% required) — 6 marks
- Active participation and progress reviews — 4 marks
- Assignment - 10 marks

3. FEATURE LIST — STUDENT

The student is the primary end user of **ICT Education**. Every feature listed below is accessible through the student-facing web portal after a secure login. The student sees only their own data — no access to other students, teachers, or administrative records.

Feature	Role	What the student can do
Profile	Student	View and update personal information, academic history, profile photo, and password.
Exam portal	Student	See upcoming exams, sit exams in-app (MCQ, written, practical coding), view marks per question after results are published.
Attendance record	Student	Check attendance by subject or date, see current percentage, receive low-attendance warnings.

Feature	Role	What the student can do
Results & transcript	Student	View grades and GPA per semester, download official PDF transcript with university seal.
Fee status	Student	See outstanding fees, download invoices, view full payment history, receive due-date reminders.
Class schedule	Student	View personal timetable by day or week, receive instant notifications when schedule changes.
Notifications	Student	Receive real-time alerts for result publishing, schedule changes, attendance warnings, and fee due dates.

4. FEATURE LIST — TEACHER

Teachers manage the academic side of the system. They create and grade exams, mark attendance, manage class schedules, and publish results through an approval workflow.

Feature	Role	What the teacher can do
Profile & courses	Teacher	Manage their own profile, view assigned courses and departments, see their full teaching history.
Exam builder	Teacher	Create exams with MCQ, short answer, descriptive, or coding questions. Assign to a class. Set marks per question. Set time limits.
Grading	Teacher	Grade submitted exams in-app, award partial marks, add written feedback per question.
Attendance marking	Teacher	Mark attendance by class and date, view per-student attendance history, generate department or semester reports.
Result submission	Teacher	Submit graded results through the approval workflow for admin sign-off before publishing.
Schedule management	Teacher	View and request changes to their timetable. Changes go to admin for confirmation.

5. FEATURE LIST — ADMIN & PARENT

Admins control the system. Parents have read-only access to their child's records through a dedicated portal.

Feature	Role	Capability
User management	Admin	Onboard, update, and deactivate student and teacher accounts without touching code.
Result approval	Admin	Review and approve submitted results before they are published to students.
Fee management (Optional)	Admin	Define fee structures per semester or department, track all payments, view real-time financial dashboard, send overdue notices.
Timetable control	Admin	Create and publish class schedules. Conflict detection prevents

Feature	Role	Capability
		double-booking of rooms or teachers.
Reports	Admin	Generate attendance, result, and fee reports by department, semester, or individual student.
Attendance summary	Parent	View child's attendance record and receive automatic alerts for absences.
Results & schedule	Parent	See published results, upcoming exam dates, and class timetable for their child.
Fee status (Optional)	Parent	Check outstanding fees and payment history. Receive overdue notifications.

6. REST API SPECIFICATION

The **ICT Education** API is a RESTful HTTP API built with FastAPI. All endpoints return JSON. Authentication uses JWT bearer tokens with role-based access control. Base URL: /api/v1

Authentication

Method	Endpoint	Description	Access
POST	/auth/login	Login, returns access + refresh token	Public
POST	/auth/refresh	Refresh access token	Public
POST	/auth/logout	Invalidate current session	All roles
PUT	/auth/password	Change own password	All roles

Users & profiles

Method	Endpoint	Description	Access
GET	/users/me	Get own profile	All roles
PUT	/users/me	Update own profile	All roles
GET	/users/students	List all students	Admin, Teacher
POST	/users/students	Create student account	Admin
GET	/users/students/{id}	Get single student	Admin, Teacher
PUT	/users/students/{id}	Update student	Admin
DELETE	/users/students/{id}	Deactivate student account	Admin
GET	/users/teachers	List all teachers	Admin
POST	/users/teachers	Create teacher account	Admin
PUT	/users/teachers/{id}	Update teacher	Admin

Exams & grading

Method	Endpoint	Description	Access
GET	/exams	List exams (filtered by role)	All roles
POST	/exams	Create new exam	Teacher

Method	Endpoint	Description	Access
GET	/exams/{id}	Get exam details and questions	All roles
PUT	/exams/{id}	Update exam	Teacher
DELETE	/exams/{id}	Delete unpublished exam	Teacher, Admin
POST	/exams/{id}/submit	Student submits answers	Student
POST	/exams/{id}/grade	Teacher grades submission	Teacher
GET	/exams/{id}/results	Get results for an exam	Teacher, Admin

Attendance

Method	Endpoint	Description	Access
GET	/attendance/me	Get own attendance summary	Student
POST	/attendance	Mark attendance for a class	Teacher
GET	/attendance/{classId}	Get attendance for a class	Teacher, Admin
PUT	/attendance/{id}	Correct an attendance record	Teacher, Admin
GET	/attendance/reports	Generate attendance reports	Admin

Results & transcripts

Method	Endpoint	Description	Access
GET	/results/me	Get own results (all semesters)	Student
POST	/results/{examId}/submit	Teacher submits results for approval	Teacher
POST	/results/{id}/approve	Admin approves and publishes results	Admin
GET	/results/{studentId}/transcript	Download transcript as PDF	Student, Admin

Fees (Optional)

Method	Endpoint	Description	Access
GET	/fees/me	Get own fee status and history	Student, Parent
POST	/fees	Define fee structure	Admin
POST	/fees/payments	Record a payment	Admin
GET	/fees/payments/{studentId}	Get payment history	Admin, Parent
GET	/fees/invoices/{id}	Download invoice PDF	Student, Admin
GET	/fees/overdue	List all overdue accounts	Admin

Scheduling

Method	Endpoint	Description	Access
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Method	Endpoint	Description	Access
GET	/schedule/me	Get own timetable	Student, Teacher
POST	/schedule	Create class schedule	Admin
PUT	/schedule/{id}	Update schedule entry	Admin
DELETE	/schedule/{id}	Remove a class from schedule	Admin
GET	/schedule/conflicts	Detect scheduling conflicts	Admin

7. WEB APPLICATION

The **ICT Education** web application is the primary interface for all users. It is a single-page application (SPA) built in React and TypeScript, served from a CDN, and communicates exclusively with the REST API described above.

Technology stack

Layer	Technology
Frontend framework	React 18 + TypeScript — component-driven, type-safe, maintainable at scale
Styling	TailwindCSS — utility-first, no design system overhead, consistent spacing and color
API communication	React Query — caching, background refetch, optimistic updates, loading and error states
Backend API	FastAPI (Python 3.12) + PostgreSQL + SQLAlchemy + Alembic for schema migrations
Authentication	JWT with role-based access (Student, Teacher, Admin, Parent) enforced at API level

Web application screens

The following screens must be delivered as part of the web application. Each screen maps directly to the API endpoints specified in Section 6.

- Login — email/password form, role-based redirect after successful authentication
- Dashboard — role-specific home screen with summary widgets (upcoming exams, attendance %, fee status, recent results)
- Profile page — view and edit personal details, profile photo, password change
- Exam list — upcoming and past exams with status badges (scheduled, open, graded, published)
- Exam room — timed exam interface supporting MCQ, written, and coding question formats
- Results view — per-subject grades, GPA, semester selector, transcript PDF download
- Attendance page — calendar or table view of attendance records with percentage indicator
- Fee centre — outstanding balance card, payment history table, invoice PDF download
- Timetable — weekly grid view of class schedule with room and teacher information
- Admin: user management — searchable table of students and teachers with create/edit/deactivate actions
- Admin: result approval — pending results queue with approve/reject workflow
- Admin: fee dashboard — real-time revenue view, overdue list, bulk notice sender
- Teacher: exam builder — question editor with type selector, marks field, hint input, and preview
- Teacher: grading interface — per-student submission view with inline marking and feedback fields
- Teacher: attendance marker — class roster with present/absent toggle per student per session

- Notifications panel — system-wide notification feed with read/unread state

8. DELIVERABLES

The following items must be delivered at project completion:

- Full source code — backend API, web frontend, database migrations — in a private Git repository
- **Video Presentation & Review**

9. CLOSING NOTE

This proposal defines a complete, production-grade university management system. Every feature described here exists because it solves a real problem for a real person inside a university. The grading structure is designed to reward working software over documentation, because the measure of this project is whether people can use it.

ICT Bangladesh

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